

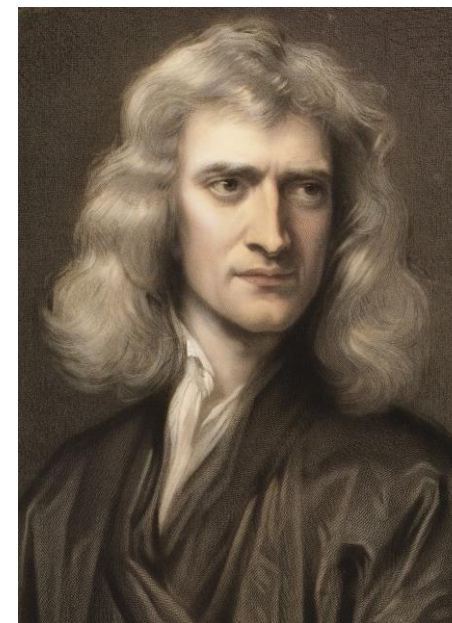
Math Primer

(4. Limits and Derivatives)

Contents



- ▶ Introduction to Limits
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- ▶ Introduction to Derivatives
- ▶ Important Results



Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)

Limits



- ▶ There are 7 undefined quantities in Maths

$$\frac{0}{0}, \infty - \infty, \frac{\infty}{0}, 0 \cdot \infty, 1^\infty, 0^\infty, \infty^0$$

- ▶ However, the above quantities are defined if the numbers are close to the above values but not exactly equal. This is the concept of limit which refers to “being in the neighborhood of “ or “tends to” a particular value.
- ▶ e.g., the following expression denotes “What is the limit of $f(x)$ when x tends to 0 (not exactly equal to 0 but as close to 0 as you can think of)”

$$\lim_{x \rightarrow 0} f(x)$$

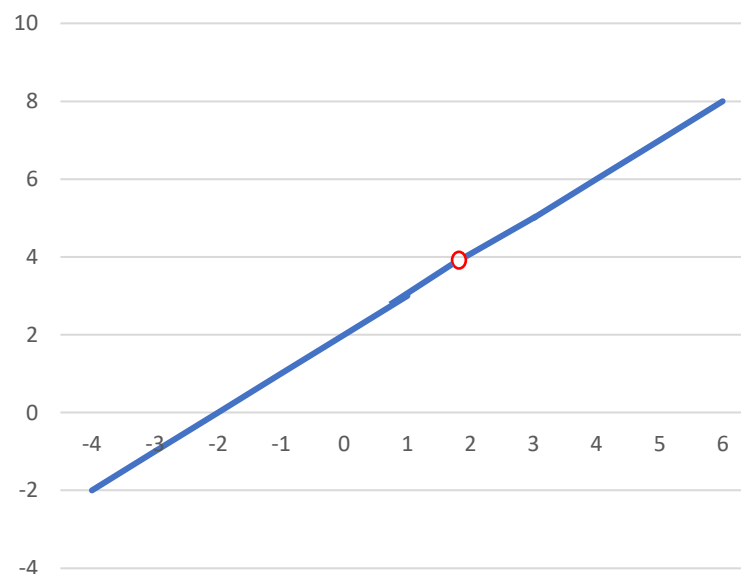
Limits



- ▶ Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}$$

- ▶ What is the domain? Its value is always $(x + 2)$, except at $(x = 2)$, where it is not defined.
- ▶ However, if x is very close to 2 but not equal to 2, then $f(x)$ will approach 4. We say, $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4$



Limits – LHL and RHL

- ▶ When we say $x \rightarrow 0$, it can approach 0 from left (we write $x \rightarrow 0^-$) or it can approach from right (we write $x \rightarrow 0^+$). So, in the first case, x will be negative and in the second case it will be positive.
- ▶ Consider the following function.

$$\underline{\underline{f(x)}} = \begin{cases} 1, & \underline{\underline{x \leq 0}} \\ 2, & \underline{\underline{x > 0}} \end{cases}$$

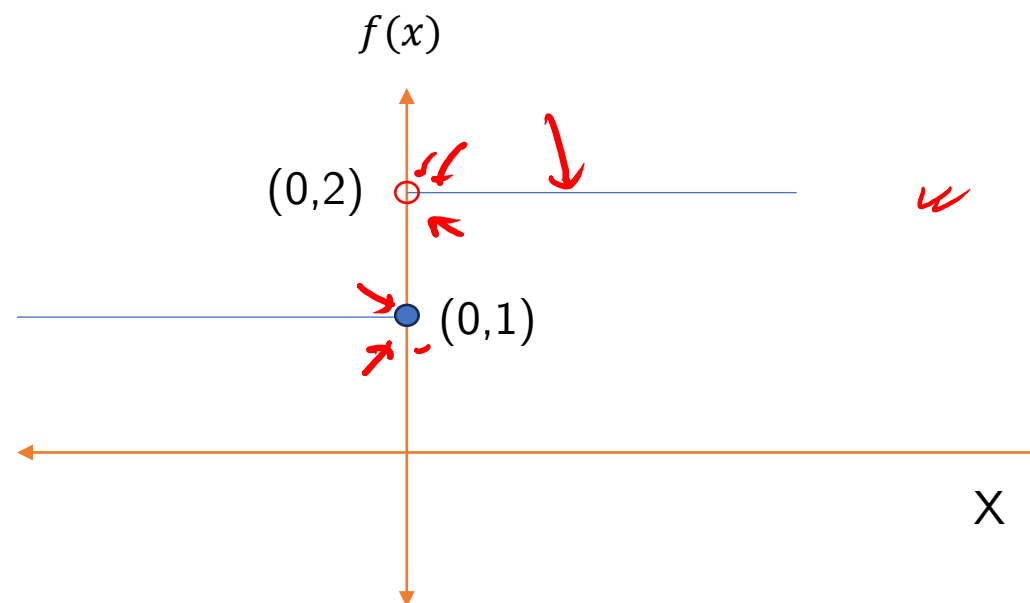
- ▶ What are the following quantities ?

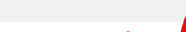
I. $\underline{\underline{f(0) = ?}}$ 1

II. $\underline{\underline{\lim_{x \rightarrow 0^-} f(x) = ?}}$ 1

III. $\underline{\underline{\lim_{x \rightarrow 0^+} f(x) = ?}}$ 2

IV. $\underline{\underline{\lim_{x \rightarrow 0} f(x) = ?}}$ limit does not exist
(x=0)



$$f(1) = 1$$
$$\lim_{x \rightarrow 1} f(x) = 1$$

$$f(1) = 2, \lim_{x \rightarrow 1} f(x) = 2$$
$$\text{LHL} = -2, \text{RHL} = 2 \quad f(0) = 0$$

Limit does not exist

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 1^-} f(x) = 3 \\ \text{RHL} &= \lim_{x \rightarrow 1^+} f(x) = 3 \end{aligned}$$
$$f(0) = 0 \quad \text{and} \quad \frac{-0.0000 \dots 1}{0.000 \dots 1} = -1$$

LHL = -1

RHL = 1

Limit does not exist.

Limits – Important Results

$$\lim_{x \rightarrow 0} (1+x)^{\overset{\infty}{1/x}} = e \quad \text{or} \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m = e^r$$

$$\rightarrow \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad \checkmark \quad \left(\frac{0}{0}\right)$$

$\log(s)$ $\ln(s)$

$$\lim_{x \rightarrow 0} \frac{\overset{\ln}{\log(1+x)}}{\underset{x}{x}} = 1 \quad \left(\frac{0}{0}\right)$$

Power Series / Taylor Series expansion.

L'Hospital's rule

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

$$\frac{e^x - 1}{x} = \boxed{1 + \frac{x}{2!} + \dots}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

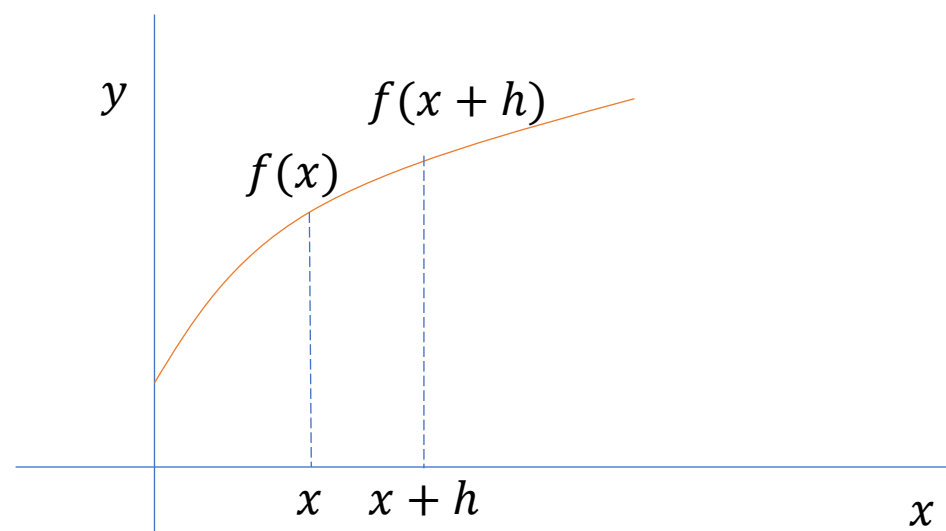
$$\boxed{1} + \frac{x}{2} + \frac{x^2}{3} + \dots$$

Derivatives

$$y = f(x)$$



- ▶ The derivative of $f(x)$ at x is defined as $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$, sometimes we write $\lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$
- ▶ The above expression is the ratio of change in $f(x)$ to change in x , when there is an **infinitesimal** change in x . In a way, derivative is the **sensitivity** of y w.r.t x
- ▶ The derivative is also the slope of the tangent to this curve at $(x, f(x))$
- ▶ It's denoted as $\frac{dy}{dx}$ or $f'(x)$... Note that ' d ' stands for **differential** whereas ' Δ ' stands for **difference**



Derivatives – A motivating Example



- ▶ Consider a square with side $x = 10$. What is the change in area of a square if the side of the square changes by (a) 5 units, (b) 1 unit , (c) 0.001 unit

Derivatives – Important Results

► Here are some of the important derivative results :

$$\rightarrow \frac{dc}{dx} = 0$$

$$\rightarrow \frac{dx^n}{dx} = \underline{nx^{n-1}}$$

$$\rightarrow \frac{de^x}{dx} = e^x$$

$$\rightarrow \frac{d \ln x}{dx} = \underline{\underline{1/x}}$$

$$\left\{ \frac{d \sin x}{dx} = \cos x \quad \checkmark \right.$$

$$\left\{ \frac{d \cos x}{dx} = -\sin x \quad \checkmark \right.$$

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{e^{x+\Delta x} - e^x}{\Delta x} &= \frac{d(e^x)}{dx} \\ &= e^x \left[\frac{e^{\Delta x} - 1}{\Delta x} \right] = e^x \cdot 1 \end{aligned}$$

Derivatives – Algebra

- Derivative is a linear operator

$$\left\{ \begin{aligned} \frac{d}{dx} [f(x) \pm g(x)] &= \frac{df(x)}{dx} \pm \frac{dg(x)}{dx} \\ \frac{d}{dx} [af(x)] &= a \frac{df(x)}{dx} \end{aligned} \right.$$

x, y
 $\alpha x \pm \beta y \rightarrow$ linear combination.
 $\frac{d}{dx} (\alpha f(x) \pm \beta g(x)) = \alpha \frac{d}{dx} (f(x)) \pm \beta \frac{d}{dx} (g(x))$
 $\frac{dy}{dx}, y', y \rightarrow$ newtonian $\left(\frac{dy}{dt} \right)$

- Product Rule

$$\frac{d}{dx} [f(x) \cdot g(x)] = f(x) \frac{dg(x)}{dx} + g(x) \frac{df(x)}{dx}$$

- Quotient Rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{df(x)}{dx} - f(x) \frac{dg(x)}{dx}}{(g(x))^2}$$

- Chain Rule

$$\left(\frac{dy}{dx} \right) = \frac{dy}{du} \frac{du}{dx} = \frac{dy}{dv} \frac{dv}{du} \frac{du}{dx}$$

$x(x)$
 $f(x)$
 $g(x) \times df(x)$
 $+ f(x) dg(x)$
 $+ df(x) \cdot dg(x)$
 dx

Derivatives – Exercises

► Find the derivatives of the following w.r.t x

I. $x^5 \rightarrow x^n - nx^{n-1}, 5x^4$

II. $e^{3x} \rightarrow$

III. $3x^2 - 6e^{-2x} \rightarrow 3 \times 2x + 12e^{-2x}$

IV. $x^3 e^{2x} \rightarrow x^3 \times e^{2x} \cdot 2 + e^{2x} \cdot 3x^2$

V. $e^{4x^2} \rightarrow e^{4x^2} \times 4 \times 2x$

VI. $\frac{e^x}{x} \rightarrow \frac{x \cdot e^x - e^x \cdot 1}{x^2}$

VII. $a^x \rightarrow x a^{x-1} \times$

$$e^x \rightarrow e^u$$

$$u = 3x$$

$$\frac{du}{dx} = 3 \rightarrow$$

$$y = e^{3x} = (e^u)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= e^u \cdot 3 = 3e^{3x}$$

$$(e^{3x}) \times 3$$

Higher Derivatives

► Till now we have seen 'first order' derivatives. We can perform 'second' and 'third' order derivatives and so on

$$y' = \frac{dy}{dx}$$

$$y'' = \underline{\underline{f''(x)}} = \frac{d^2 f}{dx^2} = \frac{d}{dx} \left[\frac{df}{dx} \right] \dots \text{Second order derivative}$$

$$\underline{\underline{f'''(x)}} = \frac{d^3 f}{dx^3} = \frac{d}{dx} \left[\frac{d^2 f}{dx^2} \right] \dots \text{Third order derivative}$$

$$f^{(n)}$$

$$f^{(4)}$$

For example, $f(x) = 4x^3$, $f'(x) = 12x^2$, $f''(x) = 24x$, $f'''(x) = 24$, $\underline{\underline{f^{(4)}(x) = 0}}$, 0 ----

Exercise : Find the nth derivative of $\underline{\underline{e^{ax}}}$

$$f'(x) = a e^{ax}$$

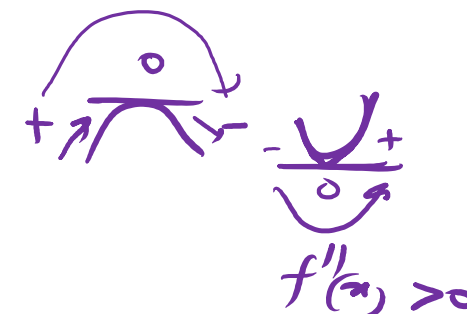
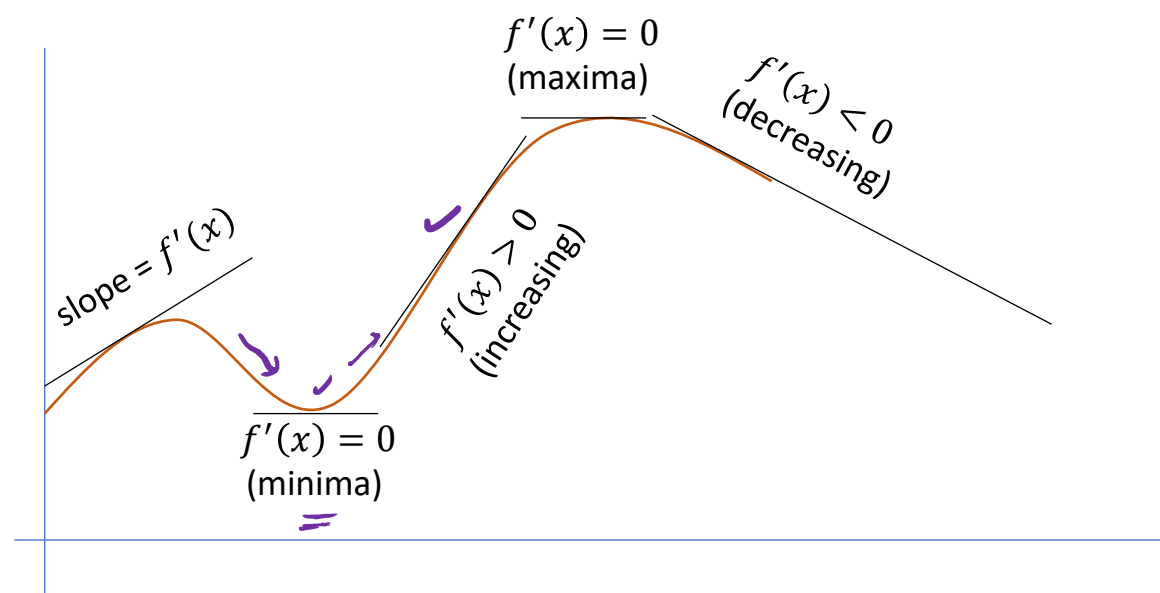
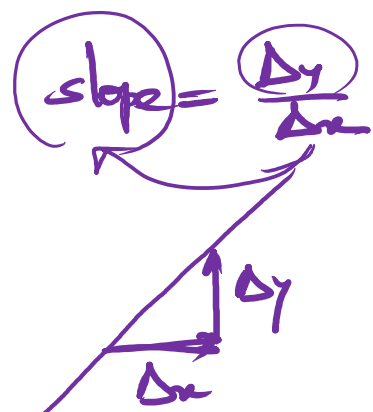
$$f''(x) = a^2 e^{ax}$$

$$f^{(n)}(x) = a^n e^{ax}$$

Application of Derivatives –

► Derivative can be used for the following applications

- I. finding the slope of the tangent to a curve (slope = $f'(x)$)
- II. To find if a function is increasing ($f'(x) > 0$) or decreasing ($f'(x) < 0$)
- III. To find maxima ($f'(x) = 0$, $f''(x) < 0$) or minima ($f'(x) = 0$, $f''(x) > 0$)



Further limits

- We can use L' Hospital's rule to find limits of quantities which appear in $\frac{0}{0}$ form or $\frac{\infty}{\infty}$ form

Example : Show the following

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad \left(\frac{0}{0} \right) = \frac{e^x}{1} = \frac{1}{1} = 1$$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \quad \left(\frac{0}{0} \right) = \frac{\frac{1}{1+x}}{1} = \frac{1}{1} = 1$$